Topic 14 — Earth & Universe

Sources baked in: NCERT Class 6–11 (Earth, atmosphere, interior, plates), PWOnlyIAS / Drishti, **Ghatnachakra** World Geography (Universe–Solar–Earth + Rocks / Volcanoes + **Earthquakes** CB–58+; **Atmosphere–Insolation–Cyclones–Humidity–Pressure–Clouds–Winds** CB–164+), UPPCS Prelims 2018–2025 **Weight:** ★★★★ — Big Bang / Chandrasekhar, rock cycle pairs, P/S waves & Richter traps, volcano status / Ring of Fire, ozone, El Niño, Coriolis, IDL / IST **Last verified:** August 2026 **Current Affairs:** Aditya-L1 (solar); Mangalyaan = first Indian Mars orbiter; Montreal vs Kyoto; El Niño–monsoon; Mauna Loa 2022 eruption note; Fukushima 2011 tsunami–nuclear note

Current Affairs (this topic)

Year	Fact	Why asked	Source
2013–14	Mangalyaan / MOM	First Indian Mars orbiter (2024)	ISRO
2023	Aditya-L1	Solar mission — not Mars	ISRO
1987 / 1997	Montreal vs Kyoto (क्योटो)	Ozone vs climate	UNEP (संयुक्त राष्ट्र पर्यावरण कार्यक्रम) / UNFCCC (यूएनएफसीसीसी)
2015	Paris Agreement (पेरिस समझौता)	NDC successor to Kyoto path	UNFCCC
2018	Science — Mars subglacial lake	2018 Q85	Science
2026	El Niño tendency vs Indian monsoon	IMD	

UKPCS Prelims 2025

Q. UKPCS Prelims 2025, Q95

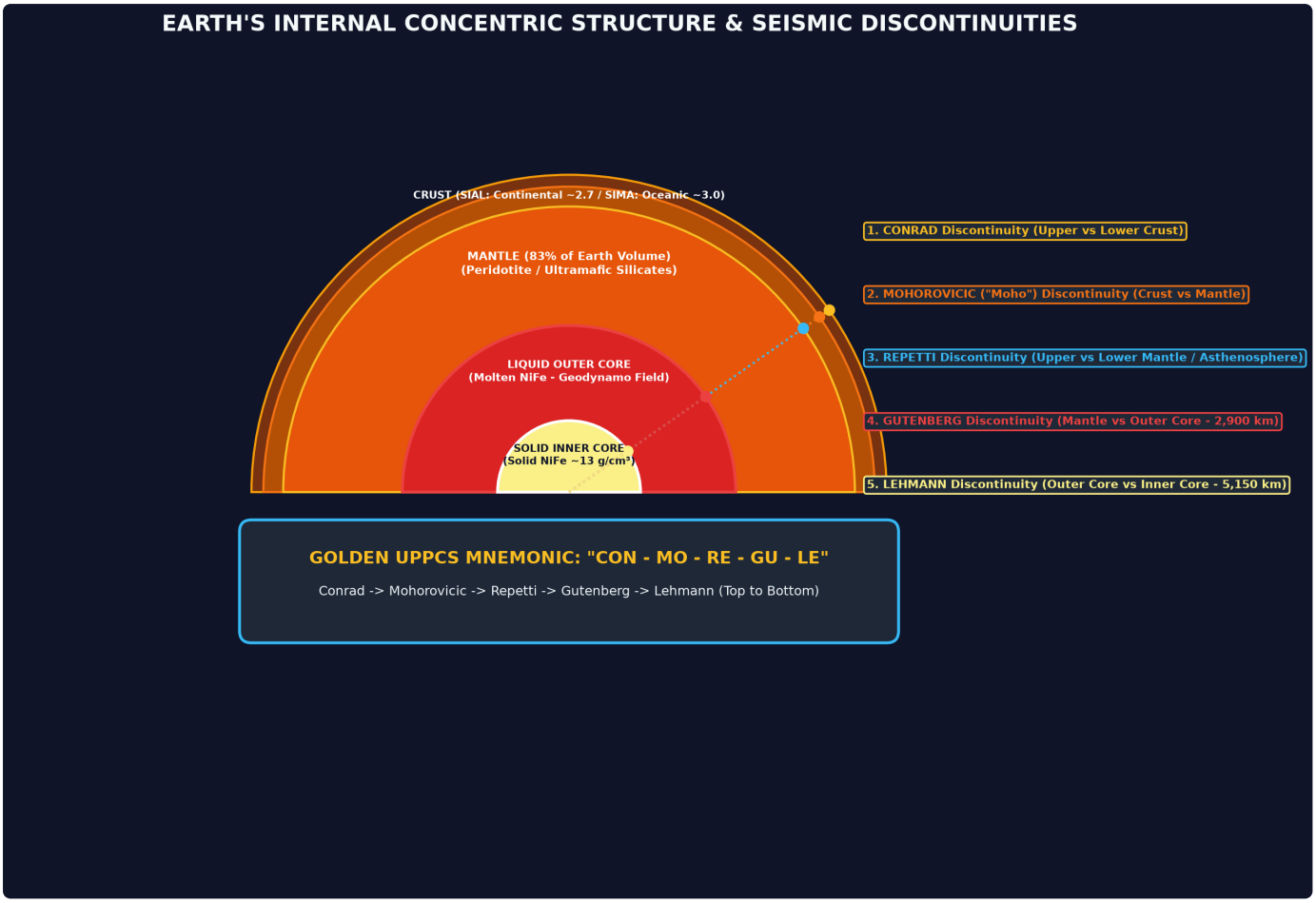
Which planet takes the longest time completing one revolution around the sun?

A. Saturn B. Uranus C. Neptune D. Jupiter

Show answer

Logic: Farthest planet among the four (चातुर्याम) has the longest year. **Ans: C (Series B provisional key).** Neptune (~165 Earth years) outranks Uranus, Saturn and Jupiter.

N.0 Framework — Earth as a System



Physical geography studies Earth as interacting spheres.

- **Lithosphere** — solid crust and upper mantle (rocks, plates, landforms).
- **Atmosphere** — gas envelope (weather, climate, winds).
- **Hydrosphere** — oceans, rivers, ice and groundwater.
- **Biosphere** — life layer where the other three meet.

Big picture order for this chapter

1. Universe and solar system set Earth's place in space.
2. Earth's motions and lat–long set day, seasons and time.
3. Atmosphere and oceans set weather–climate engines (pressure, winds, El Niño, jets).
4. Interior and plate tectonics set earthquakes, volcanoes and mountain belts.
5. Rock cycle: **igneous** (cooled magma), **sedimentary** (layers), **metamorphic** (changed by heat/pressure).

Teach the **full map** below — next year can ask any standard NCERT angle, not only the last paper's ozone or El Niño stem.

N.1 Origin of Universe

Several theories try to explain how the universe began and how the solar system formed. Learn each theory with its propounder and one clear mechanism.

Solar-system origin theories

- The **Nebular Hypothesis** of **Kant and Laplace** says a rotating gas–dust cloud flattened into a disc. The centre became the Sun and the disc condensed into planets.
- The **Planetesimal Theory** of **Chamberlin and Moulton** says a passing star pulled solar matter into space. That matter cooled into planetesimals that later grew into planets.
- The **Tidal Theory** of **Jeans and Jeffreys** says a passing star raised a gaseous filament from the Sun. The filament broke into planets.
- The **Binary-star Theory** of **Russell and Lyttleton** says a companion star was disrupted and its debris formed planets.
- **Otto Schmidt's Interstellar Dust Theory** says the Sun captured a gas–dust cloud from space and that cloud condensed into planets.

Universe-scale theories

- The **Big Bang** idea is linked to **Lemaître**, with later work often tied to **Gamow**. It starts from a hot, dense state about **13.7–13.8 billion years** ago and expands.
- The **Steady State** theory of **Hoyle, Bondi and Gold** says the universe looks the same at all times because new matter is created as space expands.
- The **Pulsating / Oscillating** model pictures repeated bang–crunch cycles of expansion and contraction.

Theory	Propounder	Teaching line
Nebular	**Kant – Laplace**	Rotating gas cloud flattened into Sun + planets
Planetesimal	Chamberlin – Moulton	Passing star pulled matter into planetesimals
Tidal	Jeans – Jeffreys	Tidal filament from a passing star broke into planets
Binary star	Russell – Lyttleton	Companion star disrupted into planetary debris
Interstellar dust	**Otto Schmidt**	Sun captured a gas–dust cloud that formed planets
Big Bang	**Lemaître** (often + **Gamow**)	Hot dense start ~**13.7–13.8 bn yr**; expansion
Steady State	**Hoyle – Bondi – Gold**	Universe looks constant; continuous creation of matter
Pulsating	Oscillating model	Repeated expansion and contraction

Galaxy, stars and distances

- The scientific study of the universe is known as **Cosmology** (*cosmos* = universe + *logos* = science).
- There are estimated to be about **100 billion galaxies (10^{11})** in the universe, and each galaxy contains on average about **100 billion stars (10^{11})**, giving an estimated **10^{22} stars** in the universe.
- The latest known close companion galaxy discovered is the **Dwarf Galaxy**.

- **Andromeda** (M31) is the nearest major spiral galaxy and the farthest celestial object visible to the naked eye. Along with the Milky Way and about 30+ other galaxies, it forms the **Local Group**, which belongs to the **Virgo Supercluster**.
- The **Milky Way** is a **barred spiral** galaxy spanning about **100,000 light-years** in diameter and about **10,000 light-years** thick at the centre, with roughly **100 billion** stars. Galileo first observed that this band of light was composed of countless individual stars.
- Only about 2,000 stars are visible to the naked eye at any given time on a clear moonless night (out of ~8,000 total visible across both hemispheres).
- One **light-year** is the distance light travels in one vacuum year at ~300,000 km/s (3×10^5 km/s or 3×10^8 m/s), equal to **9.461×10^{12} km** (or 9.461×10^{15} m).
- The size of the solar system is estimated at about **10^5 AU**.
- **Edwin Hubble** showed that galaxies recede and that recession produces **redshift** — key evidence that space expands.
- Further evidence includes the expanding universe picture and the **cosmic microwave background (CMB)**.
- The **Milky Way** is a **barred spiral** galaxy (Spitzer-era framing) with roughly **100 billion** stars in coaching notes.
- The Sun takes about **22.5–25 crore years** to orbit the galactic centre — one **galactic year**.
- An **Astronomical Unit (AU)** is the mean Earth–Sun distance, about **149.6 million km**.
- A **light-year** is the distance light travels in one year, about **9.46×10^{12} km**.
- Star **colour tracks temperature**: blue stars are hottest; red stars are cooler.
- A simple stellar life path runs from **nebula** to **main sequence** to **red giant**, then to **nova/supernova**, and finally to a **white dwarf**, **neutron star (pulsar)** or **black hole**.
- The **Chandrasekhar limit** is about **1.44 solar masses**. Below it a remnant can stay a white dwarf; above it the collapse path runs toward a neutron star or black hole (Nobel shared with Fowler, 1983).
- A **black hole** is the remnant of a massive star whose gravity is so strong that **light cannot escape**.
- The IAU lists **88** constellations. **Saptarishi / Ursa Major** points toward the **Pole Star**.
- The closest star after the Sun is **Proxima Centauri**. **Betelgeuse** is famous for Hubble surface imaging, not for being the nearest star.
- A **brittle star** is a marine echinoderm, not a cosmic object. A **byte** is a computing unit, not an astronomical measure.

Note	Fact
Expansion	**Hubble** — galaxies recede; redshift evidence
Evidence	Expanding space and the **CMB**
Galaxy	The **Milky Way** is a **barred spiral** with about **100 billion** stars
Galactic year	The Sun orbits the galactic centre in about **22.5–25 crore** years
Distance	One **AU** is about **149.6 million km**. One **light-year** is about **9.46×10^{12} km**
Star colour	Blue stars are hottest. Red stars are cooler

Note	Fact
Star life	Nebula to main sequence to red giant to nova/supernova to white dwarf / neutron star / black hole
Chandrasekhar limit	About **1.44 M⊙** is the white-dwarf ceiling
Black hole	Massive-star remnant from which light cannot escape
Constellations	The IAU lists **88**. **Saptarishi** points to the Pole Star
Closest star (after Sun)	**Proxima Centauri**
Not space	A **brittle star** is a marine animal. A **byte** is a computing unit

N.2 Solar System

Order from the Sun: **Mercury – Venus – Earth – Mars – Jupiter – Saturn – Uranus – Neptune**. **Pluto** is a **dwarf planet** since the IAU decision at Prague in **2006**.

Sun-centred model

- The modern **heliocentric** model is linked to **Nicolaus Copernicus** (*Commentariolus / De revolutionibus*).
- Coaching notes sometimes also mention an earlier Indian astronomical tradition associated with **Varahamihira** (वराहमिहिर), but Copernicus remains the standard European propounder pair for heliocentrism.
- **Kepler** gave the laws of planetary motion.
- **Newton** explained orbital motion through **universal gravitation**.

Inner and outer planets

- Inner **terrestrial** planets are Mercury, Venus, Earth and Mars — rocky worlds with few or no moons.
- Outer planets are the gas giants **Jupiter** and **Saturn** and the ice giants **Uranus** and **Neptune**.
- Earth is the third planet, densest by mean density (~**5.5 g/cm³**), tilted about **23½°**, with one moon, in the **habitable / Goldilocks** zone.
- The Sun holds about **99.8%–99.9%** of solar-system mass (the ultimate energy source for Earth). Its diameter is about **14 lakh km** (~109× Earth's) and its mass/volume is ~13 lakh times that of Earth.
- Composition of the Sun: **71% Hydrogen, 26.5% Helium, and 2.5% other elements** (H:He ratio ≈ 3:1). Its core acts as a gigantic nuclear fusion reactor converting hydrogen into helium.
- Temperature: Surface / photosphere temperature is about **6,000°C**; core temperature reaches about **15,000,000°C** (1.5 crore °C).
- The shining visible surface of the Sun is the **Photosphere** (radiates disc energy). The outer gaseous atmosphere is the **Corona** (thin hot gases), visible only during a total solar eclipse or through a special **Coronagraph** telescope.
- The Sun and its planets travel through the galaxy at about **70,000 km/h**.

- Light from the Sun (speed ~300,000 km/s) takes about **8 minutes 20 seconds** (~8.5 minutes / ~500 seconds) to reach Earth.
- A **nebula** is a star-forming cloud. It is **not** a member of the solar system family of planets, moons, asteroids and comets.

Planet desk (tilt · spin sense · moons · periods)

Most planets **orbit and rotate prograde** (same sense as Earth's west-to-east spin). **Venus** and **Uranus** are the classic **retrograde** rotators. Moon counts rise as new small moons are found — learn the stable coaching numbers and named moons first.

Planet	Axial tilt (approx.)	Rotation sense	Rotation (day)	Revolution (year)	Moons (coaching)	Must-score note
Mercury	**~0°**	Prograde	**~59** Earth days	**~88** Earth days	**0**	Smallest planet, fastest orbit, no moon
Venus	**~177°**	**Retrograde**	**~243** Earth days	**~225** Earth days	**0**	Hottest (~464°C); Morning/Evening Star
Earth	**23½°**	Prograde	**~24 h**	**365¼** days	**1** (Moon)	Densest mean; habitable zone
Mars	**~25°**	Prograde	**~24.6 h**	**~687** Earth days	**2** (Phobos, Deimos)	Red planet; day/tilt close to Earth
Jupiter	**~3°**	Prograde	**~10 h**	**~12** Earth years	Many; Galilean **Io, Europa, Ganymede, Callisto**	Largest / heaviest; fastest spin
Saturn	**~27°**	Prograde	**~10.7 h**	**~29.5** Earth years	Many; **Titan** largest of Saturn	Clearest rings
Uranus	**~98°**	**Retrograde** (sideways)	**~17 h**	**~84** Earth years	Many	Sideways rotator; Sun rises in the west framing
Neptune	**~28°**	Prograde	**~16 h**	**~165** Earth years	Many; **Triton** famous	Coldest / farthest in 8-planet set
Pluto (dwarf)	High / retrograde framing	Retrograde	Long	**~248** Earth years	Charon, Nix, Hydra and others	IAU dwarf since **2006**

Other solar-system facts

- The **largest moon in the solar system** is Jupiter's **Ganymede**, not Saturn's Titan.
- **Titan** is Saturn's largest moon and has a thick atmosphere.
- The **asteroid belt** lies between **Mars and Jupiter**.

- The **Kuiper Belt** holds icy leftovers beyond Neptune. The **Oort Cloud** is a far reservoir of comets.
- **Venus specifics**: Known as the **"Veiled Planet"** due to its impenetrable thick cloud cover (primarily sulfuric acid/CO_2); rotates clockwise (**retrograde**, east to west, like Uranus).
- **Uranus specifics**: Discovered in 1781 by **Sir William Herschel**; first planet discovered using a telescope; has **13 rings** (designated alpha, beta, gamma, delta, epsilon, lambda, etc.); axis tilt (~98°) causes it to rotate almost on its side (**"Planet on its Side"**); greenish appearance due to atmospheric **methane**.
- **Pluto & Asteroid 134340**: Demoted from planetary status by the International Astronomical Union (IAU) on **24 August 2006** at Prague; assigned minor planet / asteroid number **134340** by the Minor Planet Centre (MPC); companion moons Charon, Nix, and Hydra designated 134340 I, II, and III.
- **Trojan Asteroids**: Two clouds of asteroids that orbit the Sun along the same orbital path as Jupiter, located ~60° ahead and ~60° behind Jupiter at its Lagrange points (L4 and L5).
- **Asteroids (Planetoids)** are rocky debris between Mars and Jupiter that failed to coalesce into a planet due to Jupiter's disruptive gravitational pull. Over 5,000 have been identified, rotating every 5–20 hours.
- **Meteors and Meteorites**: Meteors ("shooting stars") are rock fragments formed by asteroid collisions that burn up in the mesosphere due to atmospheric friction. Large unburned fragments that strike the Earth are **Meteorites**.
- **Comets**: ("Hairy star", from Latin *stella cometa*); consist of frozen gases, rock, and metallic dust. When nearing the Sun, the ice vaporises into a glowing gaseous head called a **Coma**, swept back by solar radiation and solar wind into a long luminous **tail** that **always points away from the Sun**. Halley's comet returns about every **76 years**.
- Earth's albedo (~**0.3**) is higher than Mercury's (~**0.1**), so Earth reflects a larger share of sunlight even though Mercury is closer to the Sun.

Eclipses

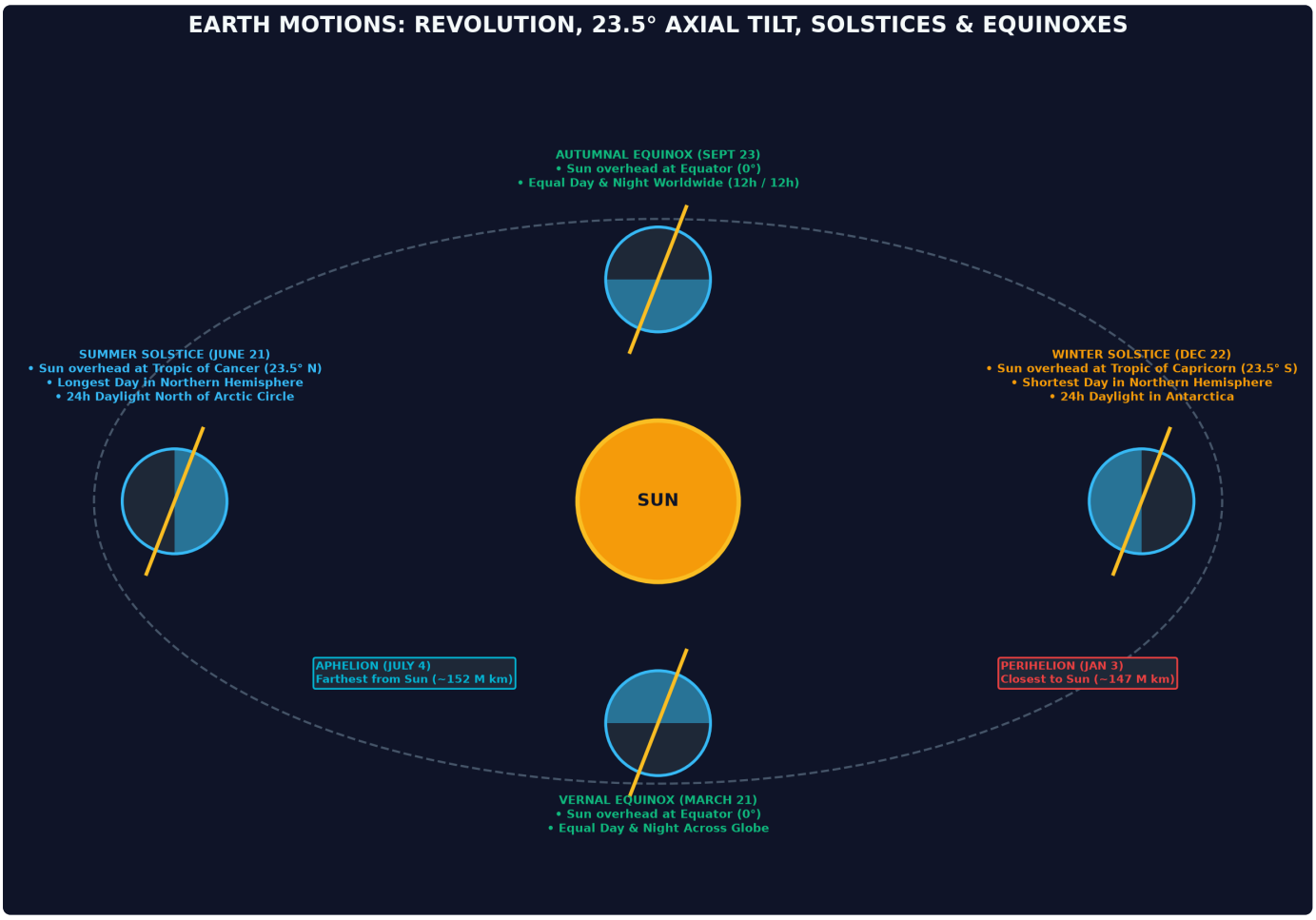
- A **solar eclipse** can occur only at **New Moon**, when the Moon lies between the Sun and Earth and the Moon's shadow falls on Earth.
- A **lunar eclipse** can occur only at **Full Moon**, when Earth lies between the Sun and Moon and Earth's shadow falls on the Moon.
- The dark central shadow is the **umbra** (total eclipse). The lighter outer shadow is the **penumbra** (partial eclipse).
- The **diamond-ring** effect appears at the start and end of totality in a total solar eclipse.

Moon facts

- The study of the Moon is called **Selenology**. The Moon is also referred to as a **fossil planet**.
- Dimensions: Diameter is **3,475 km**; circumference is **10,864 km**; mass is ~1/81 and size/volume is **1/4th** of Earth. Surface gravity is **1/6th** that of Earth.
- Distances: Minimum distance at **perigee** is **3,64,000 km**; maximum distance at **apogee** is **4,06,000 km**. Light from the Moon takes **1.3 seconds** to reach Earth.
- Periods: Rotation on its axis takes **27 days 7 hours 43 minutes** (~27.3 days, known as a **sidereal month**), which equals its revolution period around Earth (synchronous rotation). Revolution with reference to the Sun takes **29.53 days** (29d 12h 44m), known as a **synodic month** (lunar month).

- Only **59%** of the Moon's total surface is ever visible from Earth; 41% remains hidden.
- The highest mountain on the Moon is **Leibnitz Mountain** (10,660 metres), located near its south pole.
- Features: Bright areas are rugged highlands; dark plains are maria/lowlands. The **Sea of Tranquillity** (*Mare Tranquillitatis*) is a level dust plain on the Moon where **Apollo 11** landed in **1969**.
- Temperature: Extremes range from about **+100°C** during lunar day to **-180°C** during lunar night.
- The Moon has **no atmosphere, no twilight, and no sound** (sound requires a material medium to propagate).
- Surface composition: Rich in silicon, iron, and magnesium.
- **Aditya-L1** is a solar mission. Chandrayaan is a lunar programme — do not swap them.

N.3 Motions of Earth



Motion	Period	Effect
Rotation	~**24 h** (west to east)	Day and night, apparent Sun motion, and **Coriolis**
Revolution	**365¼** days	Year; leap year every 4 years (century ÷400)
Orbit shape	Ellipse	**Perihelion ~3 Jan** (closest); **Aphelion ~4 Jul** (farthest)

Seasons come from **axial tilt 23½°**, not from perihelion (NH winter is when Earth is actually *nearer* the Sun).

Date	Event	Sun overhead / polar day
21 Mar / 23 Sep	Equinox	Sun over Equator; day equals night
21 Jun	NH summer solstice	Tropic of **Cancer**; Arctic 24 h day
22 Dec	NH winter solstice	Tropic of **Capricorn**; Arctic 24 h night

Around the Northern Hemisphere **winter solstice**, global pressure belts **shift south** with the Sun. That shift helps explain winter rain on west-coast subtropical latitudes (Mediterranean-type climates).

Note	Fact
Axis to orbital plane	**66½°** (complement of 23½°)
Circle of illumination	Great circle separating day/night; tilts with seasons
Sidereal day	**~23 h 56 min** (true 360° spin) vs mean solar day **24 h**
Leap year	Extra day because revolution ≈ **365¼ d**; century years leap only if ÷ **400**

N.4 Latitude



Latitude lines are parallels. **0°** is the Equator. **90°** marks the poles. One degree of latitude is about **111 km**.

Parallel	Value	Note
Tropic of Cancer	**23½° N**	Sun overhead at NH summer solstice
Tropic of Capricorn	**23½° S**	Sun overhead at NH winter solstice
Arctic / Antarctic circles	**66½°**	90 – 23½
Heat zones	See bullets below	

Heat zones run **torrid** between the tropics, **temperate** from each tropic to 66½°, and **frigid** from there to the poles.

India on the latitude map

- India's area rank is **7th** in the world (not 6th).
- India holds about **2.4%** of world land area.
- The Tropic of Cancer runs **through the middle** of India, so India is **not** wholly tropical.
- A claim that India is split into neat equal latitudinal halves is an overclaim if asked alone.
- Uttar Pradesh (उत्तर प्रदेश) does **not** lie on the Tropic of Cancer.
- Same latitude does **not** mean same temperature. **Agra** (आगरा) and **Darjeeling** share a latitude belt idea but differ sharply because of **altitude** and thinner hill air.

Great circles, 0°–0°, and field tricks

- The intersection of **0° latitude and 0° longitude** lies in the **Atlantic Ocean** in the **Gulf of Guinea**. It is **not** on a land capital and **not** in India.
- **Great circles** include the **Equator** and **all meridians**. A parallel such as **60°N is not** a great circle.
- If the Pole Star is on your **left**, you are walking **east**.
- If stars rise **perpendicular to the horizon**, the observer is on the **Equator**.
- Counting every 1° parallel from 89°N to 89°S plus the Equator gives about **179** latitude lines.

Countries on major parallels (full lists)

Learn the **full country sets** below. Partial name chains are not enough.

Equator (0°) — 13 countries (west-to-east coaching order):

Ecuador, Colombia, Brazil, São Tomé and Príncipe, Gabon, Republic of the Congo, Democratic Republic of the Congo, Uganda, Kenya, Somalia, Maldives, Indonesia, Kiribati.

- **Not** on the Equator: **Mexico**, **Equatorial Guinea** (name trap; lies north), **Rwanda**.

Tropic of Cancer (~23½° N) — 17 countries / territories usually listed:

Mexico, Bahamas, Western Sahara, Mauritania, Mali, Algeria, Niger, Libya, Egypt (मिस्र), Saudi Arabia, United Arab Emirates, Oman, India, Bangladesh, Myanmar, China, Taiwan.

- **Not** on Cancer: **Iran** (classic wrong option).

Tropic of Capricorn (~23½° S) — 10 countries:

Chile, Argentina, Paraguay, Brazil, Namibia, Botswana, South Africa, Mozambique, Madagascar, Australia.

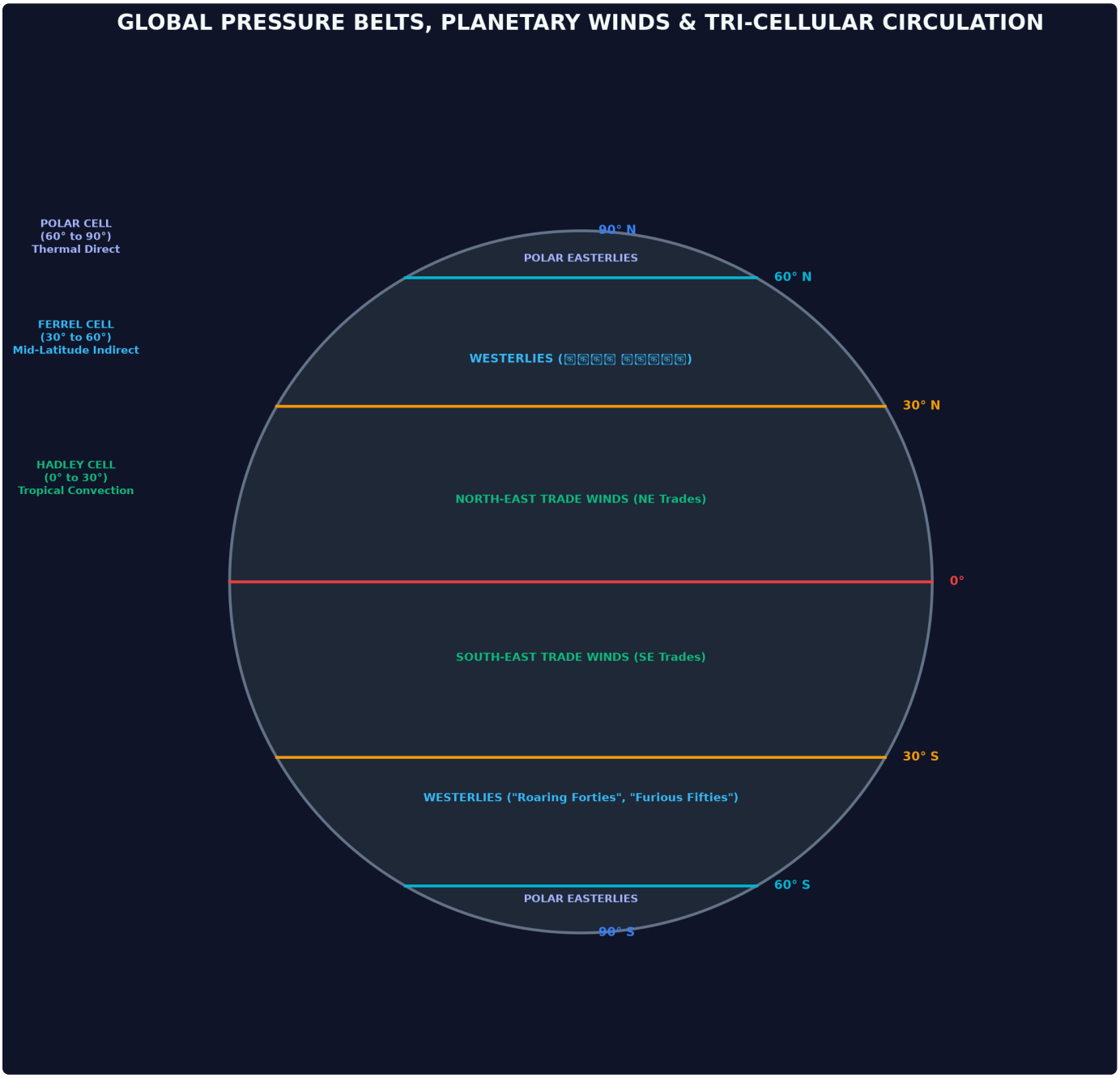
- **Not** on Capricorn: **Bolivia**.

Arctic Circle (~66½° N) — countries crossed:

Norway, Sweden, Finland, Russia, United States (Alaska), Canada, Denmark (**Greenland**), Iceland (Grímsey grazes the circle).

Antarctic Circle (~66½° S) — mainly Southern Ocean and the Antarctic continent. No crowded "capital country" list like the tropics.

N.5 Longitude, Time, Time Zones, IDL



Meridians meet at the poles. The **Prime Meridian** is **0°** longitude through **Greenwich**.

Note	Note
Earth 360° / 24 h	Roughly **15°** of longitude equals **1 hour**; **1°** equals **4 minutes**
East	Local time is **ahead** of Greenwich
IST	Based on **82°30′ E** near **Mirzapur, UP**
Time zones	Roughly 15° wide; countries zigzag for politics
IDL	Follows about **180°** (bent around islands / Aleutians)
Prime Meridian land	Full land list below — **not Niger** / Nigeria as the classic wrong option
GMT places	London, Lisbon, Accra, Dublin — **not Addis Ababa**
Cairo	About **30° E**, so about **2 h ahead** of GMT

- **IST** is **GMT + 5 h 30 min**.
- India keeps **one** time zone for the whole country.
- Crossing the IDL **westward** skips a day; crossing **eastward** repeats a day.
- The **west** side of the IDL is **ahead** (for example New Zealand versus Hawaii).

Prime Meridian (0° longitude) land countries — complete coaching list:

United Kingdom, France, Spain, Algeria, Mali, Burkina Faso, Togo, Ghana (घन).

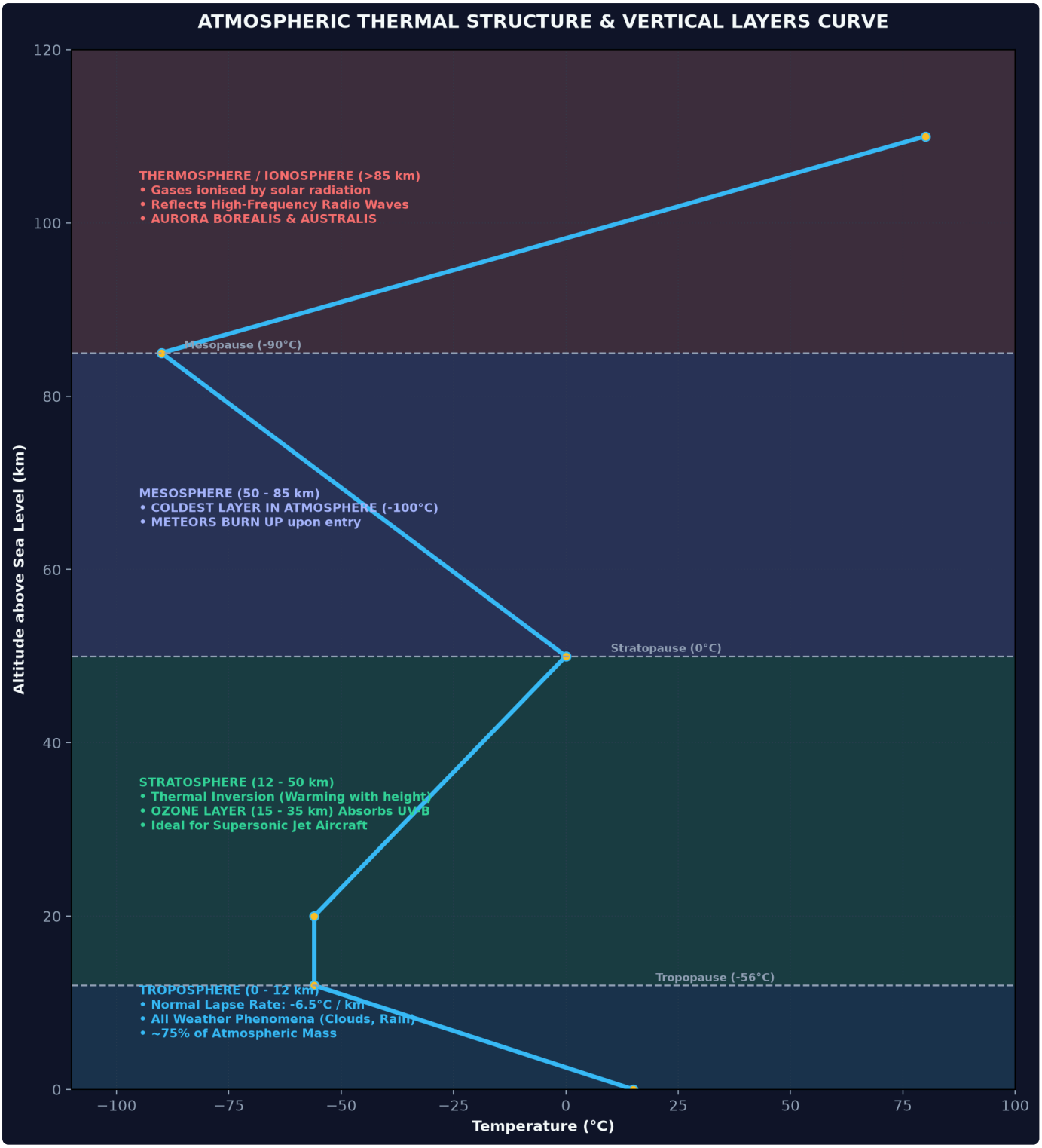
The meridian also crosses Antarctic territory, but the usual country-match set is the eight names above. **Niger** and **Nigeria** are classic wrong options.

Worked rule: Multiply the longitude difference by **4 minutes**. Add time when you move **east**. Example: **30°** east of Greenwich at Greenwich noon is **14:00**.

Country pair	Time zones
Russia	Most (11)
USA	6 (contiguous 4 + AK + HI)
China	**1** despite width
India	**1** (IST)

India’s extreme longs / Tropic **states**. **15° / IDL / IST meridian**

N.6 Atmosphere & Its Structure



The **atmosphere** is the envelope of gases held around Earth by gravity. Most of its mass sits in the lower layers. Almost all ordinary weather forms in the **troposphere**.

Composition of dry air (by volume)

Gas	Approx. share	Teaching note
Nitrogen (N_2)	**~78.08%**	Most abundant gas in dry air

Gas	Approx. share	Teaching note
Oxygen (O_2)	**~20.95%**	Needed for respiration and combustion
Argon (Ar)	**~0.93%**	Main noble gas; **not** a greenhouse gas
Carbon dioxide (CO_2) (कार्बन डाइऑक्साइड)	**~0.04%** (~400 ppm class)	Trace greenhouse gas that varies with season and human activity
Neon, helium, methane, krypton, hydrogen, etc.	Trace	Very small permanent or near-permanent fractions

Water vapour is **not** fixed in the dry-air table. Its share is **variable**. It can sit near **0%** over cold deserts and rise to about **4%** over warm humid air.

Abundance order for dry air is **nitrogen**, then **oxygen**, then **argon**, then **carbon dioxide**.

Permanent vs variable constituents

Type	Gases / constituents	Teaching note
Permanent	N_2, O_2, Ar (+ other rare noble gases)	Shares stay nearly constant day to day in the lower atmosphere
Variable	Water vapour, CO_2, O_3, aerosols / dust	Change with place, season, height and human activity

- **Argon** is common among the permanent traces, but it is **not** counted as a greenhouse gas in the usual prelims set.
- **Ozone (O_3)** is a variable gas.
- Its protective UV shield sits mainly in the **stratosphere**.
- Dust and aerosols help **scatter** light in the classic coaching set.

Vertical structure (bottom to top)

Layer	Approx. height	Temp. with height	Must-score note
Troposphere	~8 km at poles to ~18 km at the equator (avg ~13 km)	Usually **falls** (~**6.5°C/km**)	Almost all weather and most water vapour
Tropopause	Top of troposphere	Transition	Higher over the equator; lower over the poles
Stratosphere	~18–50 km	**Rises** because ozone absorbs UV	Ozone UV shield; dry and largely weather-free
Stratopause	~50 km	Transition	Boundary above the stratosphere
Mesosphere	~50–80 km	**Falls**	Coldest broad layer; meteors burn here

Layer	Approx. height	Temp. with height	Must-score note
Mesopause	~80 km	Transition	Coldest region of the atmosphere
Thermosphere / Ionosphere	above ~80 km (often framed ~80–640 km)	**Rises** strongly	Radio-wave reflection and aurora
Exosphere	outermost fringe	Merges into space	Very thin outer fringe

Correct upward order: Troposphere, then Stratosphere, then Mesosphere, then Thermosphere / Ionosphere, then Exosphere.

Layer teaching (full sentences)

- The **troposphere** is the lowest layer. It holds almost all ordinary weather and most water vapour. Temperature usually falls with height at about **6.5°C per km**.
- The **tropopause** is the lid of the troposphere. It sits higher near the equator and lower near the poles.
- The **stratosphere** holds the **ozone** shield that absorbs harmful UV. Temperature rises upward here because ozone absorbs ultraviolet energy. The layer is dry and largely free of weather, so it suits jet flight.
- The **mesosphere** cools with height. Meteors burn in this layer.
- The **thermosphere** heats strongly with height. The **ionosphere** in this region reflects radio waves and hosts the **aurora**.
- The **exosphere** is the outermost fringe that merges into space.

Homosphere (सममंडल) and heterosphere (विषममंडल)

Zone	Rough extent	Mixing
Homosphere	Surface to ~80–100 km	Turbulent mixing keeps major gas ratios roughly uniform
Heterosphere	Above ~80–100 km	Gases sort more by molecular weight

Heating, ozone and inversion

Idea	Correct
Main atmospheric heating	**Long-wave terrestrial** radiation from Earth's surface
Ozone maximum / UV shield	**Stratosphere**
Ozone hole (ओजोन छिद्र) focus	**Antarctica** in spring
Montreal Protocol (मॉन्ट्रियल प्रोटोकॉल)	Cut **CFCs** on the ozone path (not the Kyoto climate path)
Normal lapse (troposphere)	Temperature falls about **6.5°C** per km
Temperature inversion	Temperature rises with height in a short layer

- The atmosphere is heated mainly from **below**.
- Short-wave insolation warms the ground first.
- The ground then emits **long-wave** infrared, and the lower air absorbs that long-wave.
- The **ozone hole** is strongest over **Antarctica** in spring.
- The **Montreal Protocol** (मॉन्ट्रियल प्रोटोकॉल) cut CFCs that destroy stratospheric ozone (समतापमंडलीय ओजोन).
- **Kyoto** deals with climate and greenhouse gases, not the ozone protocol path.
- In a **temperature inversion**, the normal lapse is reversed for a short height band.
- Inversions are common on calm valley nights and near some fronts, and they can trap smoke and pollution near the ground.

Inline PYQ — UPPCS Prelims 2025, Q45

With reference to the ozone layer, which of the following statements is/are correct?

1. It protects the Earth's surface from solar ultraviolet radiation.
2. Its depletion has been linked to chlorofluorocarbons (CFCs) (सीएफसी).

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Logic: Statement 1 tests the UV-shield role of ozone. Statement 2 tests the CFC–depletion link. Both are true; the shield itself sits in the **stratosphere**, not the troposphere. **Ans: C.** Both statements are correct. Stratospheric ozone absorbs harmful UV. CFCs release chlorine that destroys ozone. Trap: do not place the protective ozone layer in the troposphere.

Inline PYQ — UPPCS Prelims 2022, Q138

The atmosphere is heated mainly by:

A. Incoming short-wave solar radiation alone

B. Long-wave terrestrial radiation

C. Only cosmic rays

D. Only geothermal heat

Show answer

Logic: The stem tests how the air is warmed. Incoming short-wave mainly heats the ground; the atmosphere takes most of its heat from **outgoing long-wave**. **Ans: B.** Earth's surface absorbs short-wave insolation and re-radiates long-wave infrared. Greenhouse gases and the lower air absorb that long-wave, so heating is mainly from below. Trap: "direct short-wave alone" is the usual wrong option.

N.6A Insolation · Albedo · Heat budget

The Sun is Earth's energy source. **Insolation** is incoming solar **short-wave** radiation that reaches the top of the atmosphere and the surface. Earth absorbs that energy and re-radiates **long-wave** infrared. The atmosphere is heated mainly by that **terrestrial long-wave**, which greenhouse gases trap near the surface.

What controls insolation at a place

- The **angle of the Sun's rays** matters most.
- Steeper rays concentrate energy on a smaller area.
- Slanting rays spread the same energy over a wider area, so heating is weaker.
- **Day length** rises toward the summer pole, so high latitudes can receive long hours of weak sun.
- **Atmospheric transparency** falls when dust, cloud, smoke or moisture scatter and absorb radiation.
- Earth–Sun distance changes slightly through the year, but latitude and season dominate day-to-day heating.

Maximum surface heating often sits near **~20°N**, not exactly on the equator. The equatorial belt is cloudier and wetter, so more short-wave is blocked. Clear subtropical skies let more short-wave through.

Albedo is the **reflecting** power of a surface. It is the fraction of insolation sent back to space.

Surface	Typical albedo
Fresh snow	**75–95%**
Deserts	**20–30%**
Prairie (प्रेयरी) / grassland	**10–20%**
Paddy / dark wet fields	**3–15%**
Earth mean (planetary)	**~30%**

Earth's mean albedo is about **0.3**. Mercury's mean albedo is about **0.1**. Earth therefore reflects a larger share of sunlight even though Mercury is closer to the Sun.

Heat budget (NCERT 100-unit picture)

Of **100** units of insolation at the top of the atmosphere:

Fate	Units
Reflected / scattered back to space (**planetary albedo**)	**~35**
Absorbed by the atmosphere	**~14**
Absorbed by Earth's surface	**~51**

- The surface returns that energy as **long-wave** radiation, sensible heat, and latent heat from evaporation.

- Water vapour absorbs much **infrared**.
- Cloudy nights stay warmer because clouds reflect and trap outgoing long-wave.
- On cloudy nights the ground may not cool enough for **dew**.
- Tropical cyclones weaken inland because they lose the ocean's **latent-heat** fuel once they leave warm water.

N.6B Humidity · Dew

Humidity is the amount of water vapour in the air. That vapour comes mainly from **evaporation** over oceans, lakes, wet soils and plants.

Humidity terms — Teaching Card

- **Absolute (निरपेक्ष) humidity** (निरपेक्ष) is the **mass of water vapour** in a given **volume** of air — how many grams of vapour sit in a cubic metre.
- **Relative (सापेक्ष) humidity** (सापेक्ष आर्द्रता) is that vapour expressed as a **percentage of saturation** at the **same temperature** — how full the air is compared with what it could hold.
- **Specific humidity** is the mass of vapour per unit mass of **moist air** — useful because it stays steadier when air expands or contracts.
- Relative humidity **falls as temperature rises** if the vapour mass stays the same, because warmer air can hold more water before saturation.
- Moisture generally **decreases with latitude**.
- About **90%** of atmospheric moisture sits in the lowest ~**5 km**.
- Summer "humid heat" feels **muggy** when absolute vapour is high and evaporation from the skin is slow.

Term	Quick recall
Absolute humidity	Mass of vapour per volume of air
Relative humidity	% of saturation at that temperature
Specific humidity	Mass of vapour per mass of moist air

Dew and frost

- **Dew** forms when a clear, calm night lets the ground lose long-wave heat until air near the surface cools to the **dew point**.
- Vapour then condenses on cold surfaces as liquid dew.
- Cloudy nights keep the surface warmer, so dew often fails.
- If the surface temperature drops below **0°C**, the same process yields **frost** instead of liquid dew.

N.6C Clouds

A **cloud** is a visible mass of tiny water droplets, ice crystals, or both, suspended in the air. Clouds form when moist air cools to **saturation** and water vapour **condenses** on condensation nuclei such as dust, salt (लवणाध्यक्ष) or smoke particles.

How clouds form

- Rising air cools by **expansion**. That is why cumulus towers grow on hot afternoons.
- Air can also cool when it moves over a colder surface, or when it is forced up a mountain slope.
- Condensation needs nuclei. Clean air without particles does not form cloud drops easily.
- Most ordinary clouds live in the **troposphere**, where almost all weather sits.
- Clouds are named by **height** and by **form**.

Height families

Family	Typical height (mid-latitudes)	Make-up	Teaching note
High	About **5–13 km**	Mostly **ice crystals**	Names often begin with **cirro-**
Middle	About **2–7 km**	Water drops and/or ice	Names often begin with **alto-**
Low	Surface to about **2 km**	Mostly water drops	Includes stratus sheets and many rain clouds
Vertical / towering	Base low; top may reach the tropopause	Strong convection	**Cumulus** and **Cumulonimbus** grow upward through layers

Heights are lower in polar air and higher in tropical air. The names stay the same.

Form families

Form idea	Look	Classic members
Cirriform	Thin, wispy, fibrous	Cirrus, cirrostratus, cirrocumulus
Stratiform	Sheet or layer	Stratus, altostratus, nimbostratus
Cumuliform	Heap / cauliflower	Cumulus, cumulonimbus, altocumulus
Nimbus	Rain-bearing	**Nimbostratus**, **Cumulonimbus**

- **Nimbus** in a name signals rain.
- **Cirro-** points to high icy cloud.
- **Alto-** points to middle-level cloud.
- **Stratus** means a sheet. **Cumulus** means a heap.

Main cloud types

Cloud	Family	Teaching note
Cirrus	High	Thin, wispy ice trails. Often fair-weather or approaching weather aloft
Cirrostratus	High	Thin icy sheet. May make a Sun or Moon **halo**
Cirrocumulus	High	Small icy ripples or “mackerel sky” patches
Altocumulus	Middle	White or grey patches and rolls in the middle troposphere
Altostratus	Middle	Grey sheet that can dull the Sun. May thicken before steadier rain
Stratus	Low	Grey layer hugging the weather layer. Light drizzle or dull sky
Nimbostratus	Low	Thick rain sheet. Steady, often widespread rain or snow
Cumulus	Vertical	Puffy cauliflower heaps. Isolated fair-weather cumulus means unsettled tall storm not yet formed
Cumulonimbus	Vertical	Towering storm cloud. Intense rain, hail, thunder and lightning

Cloud teaching (full sentences)

- **Cirrus** clouds are the highest common clouds. They are made of ice and look like white feathers or threads.
- **Cirrostratus** can spread as a thin veil. A bright ring or **halo** around the Sun or Moon often means cirrostratus is present.
- **Cirrocumulus** forms small high ripples. It is less common than cirrus in ordinary day-to-day weather talk.
- **Altocumulus** sits in the middle troposphere as patches or rolls.
- **Altostratus** is a mid-level grey sheet. The Sun may look watery through it.
- **Stratus** is a low grey blanket. It can give drizzle or a dull, overcast day.
- **Nimbostratus** is the classic steady-rain cloud. Rain from it is usually continuous rather than brief and violent.
- **Cumulus** looks like cotton heaps with flat bases. Fair-weather cumulus often forms by afternoon heating and then dies by evening.
- **Cumulonimbus** is the thunderstorm cloud. Strong convection builds it into a tall tower that can bring intense rain, hail and thunder.
- Isolated fair-weather cumulus can grow into cumulonimbus when heating and moisture keep feeding the updraft.

Fog, halo and thunder links

Idea	Correct
Fog	Cloud with its base at or very near the **ground**
Halo around Sun / Moon	Often **cirrostratus**

Idea	Correct
Steady widespread rain	Often **nimbostratus**
Intense convective rain / hail / thunder	**Cumulonimbus**
Fair-weather afternoon heaps	Isolated **cumulus**

- **Fog** forms when air near the ground cools to the dew point and condenses. It is cloud at ground level, not a separate gas.
- **Thunder** is the sound of air expanding after a lightning flash heats it in a fraction of a second.
- Thunder is **not** caused only by "two clouds meeting."
- Lightning and thunder belong with **cumulonimbus**, not with thin high cirrus alone.

Cloud–weather quick pairs

- High icy cirrus or cirrostratus may appear first when a weather system is still far away.
- Thickening altostratus often means steadier rain is approaching.
- Nimbostratus brings the long rainy spell.
- Cumulonimbus brings the short, violent storm, often with thunder.
- An anticyclone's sinking air suppresses tall cloud growth, so skies stay clearer.

N.6D Tropical cyclones · Tornado · Anticyclone

Tropical cyclone

A **tropical cyclone** is a warm-core storm that spirals into a deep **low-pressure** centre over warm tropical seas.

- In the **Northern Hemisphere** the surface circulation is **anticlockwise**.
- In the **Southern Hemisphere** it is **clockwise**.
- Moist air rises in spiral rainbands. In a mature storm, air **descends** in the calm **eye**.
- The eye column is **warmer** than the surrounding cloud wall aloft — do not mark "eye is 10°C colder."
- Fuel is warm ocean water and **latent heat**. The storm weakens over land because that ocean fuel is cut off.
- Only some cyclones develop a clear eye. Jet streams exist in **both** hemispheres and can steer or shear storms.

Local name	Region
Hurricane	USA / Atlantic–Caribbean
Typhoon	China Sea / western Pacific
Willy-Willy	North-west Australia
Cyclone	North Indian Ocean / India

Local name	Region
Baguio	Philippines
Taifu	Japan

- A **tornado** is a smaller, intense funnel. The classic USA belt is **Tornado Alley** (Texas–Oklahoma–Kansas–Nebraska plains). Intensity uses the **Fujita** scale. Hurricane damage uses **Saffir–Simpson**.
- Cyclone **Nargis** hit **Myanmar**, not India.
- Tropical cyclones are rare in the South Atlantic and south-east Pacific tropics mainly because **sea-surface temperature stays too low**.

Anticyclone

An **anticyclone** is a **high-pressure** centre.

- Air **sinks** and then **outflows** at the surface.
- In the Northern Hemisphere the outflow is **clockwise**.
- In the Southern Hemisphere the outflow is **anticlockwise**.
- Fair, settled weather is common under an anticyclone because sinking air suppresses cloud growth.
- A sudden **fall of mercury** in the barometer indicates an approaching storm/cyclone; a slow rise indicates fair, clear weather; a slow fall indicates rain.
- **Atmospheric pressure units**: Standard sea-level pressure is **76 cm of Hg** = **760 mm of Hg** = **29.92 inches of Hg** = **1013.25 mb** (millibars, or hectopascals hPa).
- **Meteorological measuring instruments**:
- **Mercurial Barometer (Fortin's Barometer)** / **Aneroid Barometer**: Atmospheric pressure.
- **Barograph**: Continuous self-recording aneroid barometer.
- **Altimeter**: Altitude measurement using atmospheric pressure variation (used in aviation).
- **Wind Vane / Weathercock**: Wind direction.
- **Anemometer**: Wind velocity / speed.
- **Hygrometer / Psychrometer**: Relative humidity of air.
- **Hyetograph**: Automatic self-recording rain-gauge.

System	Centre	Surface spin (NH)	Surface spin (SH)	Typical weather
Cyclone / low	Low	Anticlockwise	Clockwise	Cloud, rain, strong wind
Anticyclone / high	High	Clockwise	Anticlockwise	Clear, settled, light wind

N.7 Weather vs Climate

Weather is the state of the atmosphere over a short spell — hours to a few days — at a place. **Climate** is the long-term average of weather, usually taken over about **30 years**, plus the typical seasonal pattern.

	Weather	Climate
Time	Hours–days	**~30-year** average (WMO defines classical period as 30 years; coaching notes often cite 30–35 years)
Example	Today's rain, today's Loo (लू)	Monsoon climate, Mediterranean

Important Geographical Isopleths (Lines of Equal Value)

Line (Isopleth)	Parameter / Value
Isobars	Equal atmospheric pressure
Isotherms	Equal temperature
Isohyets	Equal rainfall / precipitation
Isohalines	Equal ocean / water salinity
Isobaths	Equal depth below water/sea surface
Isohypse / Contours	Equal elevation above mean sea level
Isohels	Equal amount of sunshine / sunshine duration
Isonif	Equal amount of snowfall / snow
Isorymes	Equal frost
Isobronts	Places experiencing a thunderstorm at the same time
Isochrones	Places located at equal travel time from a common centre
Isogeotherms	Equal subterranean / underground temperature
Isoseismals	Equal earthquake shaking intensity
Isogonic lines	Equal magnetic declination (Agonic line = zero declination)

Both share the same **elements**: temperature, pressure, wind, humidity, cloudiness and precipitation. Weather answers "what is happening now." Climate answers "what is usual for this season and place."

Mediterranean-type climate (Cs) — world map, not one coast only

A Mediterranean climate has **mild, wet winters** and **hot, dry summers**. Rain falls mainly when the **westerlies** migrate over the coast in the cool season. Summers stay dry under the subtropical high.

Why winter rain happens: pressure belts **follow the Sun**. Around the NH **winter solstice** the belts shift **south**, so the rain-bearing westerlies cover about **30°–45°** west coasts. In summer the subtropical high sits over those latitudes and rain is scarce.

Where else it occurs (same Cs logic):

- Classic **Mediterranean Basin** — Spain, southern France, Italy, Greece, coastal North Africa, Levant (लेवेंट) fringe.

- **California** west coast (USA).
- Central **Chile**.
- **South-west Australia** (Perth belt) and parts of southern Australia.
- **Cape region** of South Africa.
- Parts of central Chile / California are the Southern / Northern Hemisphere mirrors of the same west-coast subtropical pattern.

Do not freeze the idea as "only the Mediterranean Sea." Any similar west-coast subtropical latitude can show the same dry-summer / wet-winter rhythm.

Inline PYQ — UPPCS Prelims 2024, Q79

With reference to the Mediterranean region, which of the following statements is/are correct?

1. It receives rainfall mainly in winter.
2. Around the winter solstice, pressure belts shift southward.

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Logic: Statement 1 tests the classic Mediterranean rainfall season. Statement 2 tests the solstice shift of pressure belts. Both are true together. The same belt-shift logic also explains California, Chile, SW Australia and Cape Town-type climates. **Ans: C.** Mediterranean coasts get most rain in **winter**. Pressure belts follow the Sun and shift **south** around the NH winter solstice. Trap: do not swap this with monsoon summer rain.

N.8 Global Warming

The **natural greenhouse effect** (प्राकृतिक ग्रीनहाउस प्रभाव) keeps Earth warm enough for life. Greenhouse gases let short-wave insolation in but trap **outgoing long-wave** radiation. **Global warming** is the extra warming from a **rise in greenhouse-gas concentrations** after industrial fossil-fuel use, deforestation and intensive agriculture.

Is a greenhouse gas	Not a greenhouse gas
Water vapour, **CO_2, CH_4, N_2O, O_3, CFCs / HCFCs**	**Argon**
Natural coaching set often lists CO_2, CH_4, O_3 and N_2O	"Nitrogen oxide" (NO / NO_2) in the 2020 key is **not** treated as that natural GHG set

- Water vapour is the strongest natural greenhouse gas by abundance.

- **CO_2** is the main long-lived driver of recent warming in policy and coaching keys.
- **Methane** and **nitrous oxide** are potent even at lower volumes.
- **Argon** is an inert permanent gas. It is **not** a greenhouse gas.

Climate and ozone treaties (do not swap)

Instrument	Year	Focus	What it does
Montreal Protocol	**1987**	**Ozone layer**	Cuts ozone-depleting substances, especially **CFCs** and related chemicals
Kyoto Protocol	**1997**	**Climate / GHG**	Sets binding emission cuts for listed greenhouse gases and frames **carbon credits**
Paris Agreement (पेरिस समझौता)	**2015**	**Climate / NDCs**	Each country files Nationally Determined Contributions under the UNFCCC path

Montreal Protocol (1987) (मॉन्ट्रियल प्रोटोकॉल)

- Montreal is an **ozone** treaty, not a climate-mitigation treaty.
- It targets CFCs and other ozone-depleting substances that destroy stratospheric ozone.
- The ozone shield itself sits mainly in the **stratosphere**.

Kyoto Protocol (1997) (क्योटो प्रोटोकॉल)

- Kyoto is a **climate** treaty under the greenhouse-gas path.
- It deals with CO_2 and other GHGs, not with “saving water.”
- The idea of **carbon credit** (कार्बन क्रेडिट) trading is classically linked with Kyoto.

Paris Agreement (2015) (पेरिस समझौता)

- Paris continues the climate path after Kyoto.
- Countries submit **Nationally Determined Contributions (NDCs)** stating how they will cut or control emissions.
- Paris is still about climate and GHGs. It is not an ozone protocol and not a wetland or wildlife trade (पण्याध्यक्ष) treaty.

Common traps

Wrong pairing	Correct
Kyoto → save water	Kyoto → climate / GHG / carbon credit
Montreal → climate mitigation (जलवायु शमन)	Montreal → ozone / CFCs
Ramsar (रामसर) → climate mitigation	Ramsar → wetlands
CITES (साइट्स) → climate mitigation	CITES → wildlife trade
Paris → ozone CFCs	Paris → climate NDCs

Montreal protects the ozone layer. Kyoto and Paris address climate greenhouse gases. Do not swap those families.

Inline PYQ — UPPCS Prelims 2019, Q132

Which of the following is NOT correctly matched?

A. Cartagena Protocol — Biosafety

B. Stockholm Convention — Persistent organic pollutant

C. Montreal Protocol — Ozone layer

D. Kyoto Protocol — Save water

Show answer

Logic: The stem asks for the **wrong** pair. Three protocols are matched to their true themes. Kyoto is climate / GHG, so “save water” is the false match. **Ans: D.** Kyoto deals with climate change (जलवायु परिवर्तन) and greenhouse gases, including the carbon-credit idea. It does **not** deal with saving water. Trap: Montreal correctly pairs with the ozone layer, so do not mark C.

Inline PYQ — UPPCS Prelims 2024, Q35

Which of the following international agreements is related to climate change mitigation?

A. Montreal Protocol

B. Ramsar Convention

C. Kyoto Protocol

D. CITES

Show answer

Logic: The stem asks which agreement is about **climate-change mitigation**. Montreal is ozone. Ramsar is wetlands. CITES is wildlife trade. Only Kyoto is the climate / GHG treaty among the options. **Ans: C.** Kyoto targets greenhouse-gas mitigation. Montreal protects ozone. Ramsar protects wetlands. CITES regulates wildlife trade. Trap: Montreal is the usual wrong climate option.

Inline PYQ — UPPCS Prelims 2021, Q125

The concept of 'carbon credit' originated from

A. Earth Summit, Rio-de-Janeiro

B. Kyoto protocol

C. Montreal protocol

D. G-8 Summit, Heiligendum

Show answer

Logic: The stem asks where the **carbon credit** idea comes from. That framing belongs to the Kyoto climate path, not Montreal ozone control. **Ans: B.** Carbon credits are linked with the **Kyoto** climate framework. Montreal is the ozone / CFC treaty, so it is the common distractor.

Among major emitters, coaching ranks often run **China > USA > India** (India about third). Do not freeze any one year's option band as India's permanent rank. Warmer oceans and a warmer atmosphere favour more intense cyclones, storms and heat extremes. Nuclear power and geothermal heat are **not** forms of stored solar energy. Biomass and wind **are** forms of solar-driven energy in the usual classification.

N.9 El Niño · La Niña (ला नीना) · ENSO

ENSO (El Niño–Southern Oscillation) is the coupled ocean–atmosphere swing of the **equatorial Pacific**. The **Southern Oscillation** is the **Darwin–Tahiti** pressure see-saw.

Normal (neutral) Walker pattern

- Trade winds blow **west** across the tropical Pacific.
- Warm surface water piles up in the **western Pacific** (Indonesia–Australia warm pool).
- Cool water **upwells** off **Peru / Ecuador** in the eastern Pacific.
- Upwelling brings nutrients, so plankton and the Peru fishery stay rich.
- The **Walker cell** rises in the west and sinks in the east.

El Niño

- Trade winds **weaken** or reverse.
- Warm water spreads into the **eastern Pacific**. A warm current appears off **Peru**.
- Eastern upwelling **weakens**, so **plankton and fish catch fall** — the warm water does **not** increase Peruvian plankton.
- Heavy rain and floods can hit the normally dry west coast of South America.
- Drought (सूखा) risk rises over parts of **Indonesia, Australia** and sometimes **India's monsoon** timing/strength.
- The Walker cell **breaks or reverses**.
- A **strong / "super" El Niño** is simply an intense version of the same eastern warming and upwelling collapse. Recent strong events (for example **2015–16** and **2023–24**) are living CA reminders — teach the mechanism, not only one year's headline.

La Niña

- Trade winds are **stronger** than normal.
- Eastern Pacific stays **cooler**; upwelling **strengthens**.
- Plankton and fish often **rise** off Peru.
- Australia / Indonesia can be wetter; India's monsoon response varies by event.
- The Walker cell is **strong**.

	El Niño	La Niña
E. Pacific SST	**Warm** off **Peru**	Cool
Upwelling	**Weakens**	Strengthens
Plankton / fish	**Fall**	Rise
Walker cell	Breaks / reverses	Strong
Typical risk sketch	E. Pacific rain; Indo–Aus drought risk	Stronger trades; cooler east Pacific

Inline PYQ — UPPCS Prelims 2023, Q67

With reference to El Niño, which of the following statements is/are correct?

1. It is associated with a warm current off the Peru coast in the eastern Pacific.
2. That warm water increases plankton growth off Peru.

A. 1 only

B. 2 only

C. Both 1 and 2

D. Neither 1 nor 2

Show answer

Ans: A — Statement 2 is false. Less upwelling → **less** plankton/fish.

N.10 Coriolis Force

Coriolis force arises because Earth **rotates** west to east. It does not create wind by itself. It only **deflects** air and water that are already moving relative to the rotating Earth.

Under **Ferrel's law**, deflection is to the **right in the Northern Hemisphere** and to the **left in the Southern Hemisphere**. The effect is **zero at the equator** and grows toward the poles. That is why tropical cyclones cannot form on the equator: without Coriolis, air rushes straight into a low and does not spin into a storm.

Coriolis shapes the **trade winds**, **westerlies** and the sense of spiral in cyclones and anticyclones. It also bends **ocean currents**. It is **not** caused by Earth's magnetic field. Temperature and pressure differences start the motion; Coriolis only steers that motion.

Inline PYQ — UPPCS Prelims 2023, Q68

Winds deflect to the left in the Southern Hemisphere because of:

A. Temperature difference alone

B. Coriolis force

C. Magnetic field

D. Pressure alone

Show answer

Logic: The stem asks why Southern Hemisphere winds turn left. The key is Coriolis / Ferrel's law, not magnetism or pressure alone. **Ans: B.** Southern Hemisphere deflection is **leftward** under Coriolis (Ferrel's law). Trap: magnetic field is a common distractor; pressure starts motion but does not set the left/right rule.

N.11 Jet Stream

Jet streams are narrow ribbons of very fast wind in the upper troposphere, often near the **200–300 hPa** level. They form where strong temperature contrasts create steep pressure gradients aloft.

The main **polar jet** and the **subtropical westerly jet (STWJ (पश्चिमी जेट))** both blow from **west to east**. Speeds are often **300–500 km/h**. Pilots and WWII meteorology first mapped these cores as high-altitude **westerlies**, not as a global easterly belt.

Over India, the subtropical westerly jet sits south of the Himalaya in winter and helps winter western disturbances. In the southwest monsoon season that jet weakens or shifts, and a **Tropical Easterly Jet (TEJ)** appears near about **14°N** in the upper troposphere. TEJ is the important **easterly** exception; it must not be used to call all jets easterly.

Inline PYQ — UPPCS Prelims 2024, Q60

Assertion (A): Jet streams discovered in WWII are high-altitude **easterly** winds. Reason (R): Jet streams flow at about 300–500 km/h.

A. Both A and R are true and R explains A

B. Both A and R are true but R does not explain A

C. A is true but R is false

D. A is false but R is true

Show answer

A/R logic: A claims WWII jets were easterlies — false for the main jets. R's speed band is true but does not fix A's direction error. **Ans: D.** Main jets are **westerly**; TEJ is the easterly Indian-summer exception. The **300–500 km/h** speed figure is true, so R stands alone.

N.12 Pressure Belts, Planetary Winds, ITCZ

Pressure belts **follow the Sun**. They shift **north** in Northern Hemisphere summer and **south** around the Northern Hemisphere winter solstice.

Latitude	Belt	Wind
0°	Equatorial low / **ITCZ**	Doldrums with rising air and rain
30°	Subtropical **high**	Horse latitudes; **Trade winds** (NE/SE) blow toward ITCZ
60°	Subpolar **low**	**Westerlies**
90°	Polar **high**	Polar easterlies

Doldrums and horse latitudes

- The **Doldrums** are the calm, light-wind zone of the **ITCZ** near the Equator. Air rises here, so surface winds are weak and rainfall is frequent.
- The **ITCZ is a low**, not a high.
- **Horse latitudes** are the calm belts under the subtropical **highs** near about **30°–35°** north and south. Air sinks here, so skies are often clear and dry.
- Do not mark horse latitudes as a low-pressure belt.
- A **monsoon** is a seasonal land–sea wind reversal. It is **not** one of the permanent planetary pressure belts, even though the ITCZ shift helps drive the Indian monsoon trough.

Named Southern Hemisphere westerlies

Named belt	Fact
Roaring Forties	About **40°S**
Furious Fifties	About **50°S**
Shrieking / Screaming Sixties	About **60°S**

- Southern Hemisphere westerlies are stronger because there is less land to block them.
- **Wellington** lies in the path of the Roaring Forties.
- Trade winds help warm the **western** tropical oceans. Temperate **westerlies** help warm the **eastern** sides of ocean basins in mid-latitudes.

Local mountain winds

- A **valley breeze** is **anabatic** (आरोही) — day air flows **upslope**.
- A **mountain breeze** is **katabatic** (अवरोही पवन) — night air flows **downslope**.

N.13 Interior of the Earth

Direct drilling barely scratches the crust, so the deep interior is known mainly from **seismic waves**, density and magnetism. Body waves change speed and path at major boundaries. Those jumps mark the shells below.

Boundary	Depth pair	Fact
Moho	Base of crust (~8–40 km)	Crust / mantle
Gutenberg	**~2900 km**	Mantle / **outer core — S-waves stop** (liquid)
Lehmann	~5100 km	Outer / **inner core** (solid iron-nickel)
Layer chemistry (names)	Fact	
---------- ------	------	
Sial	Continental crust (~2.7–3.0 g/cm³) — silica + aluminium; granite; less dense than sima	
Sima	Oceanic crust — silica + magnesium; basalt; denser than sial	
Nife	Core (~13+ g/cm³ at centre) — nickel + iron (thickness ~3,471 km, core temp ~5000°C)	

Whole Earth vs Crust Chemical Composition (by weight): - **Whole Earth Composition**: **Iron (Fe) ~35%** > **Oxygen (O) ~30%** > **Silicon (Si) ~15%** > **Magnesium (Mg) ~13%** > Nickel (Ni) ~2.4% > Sulphur (S) ~1.9% > Calcium (Ca) ~1.1% > Aluminium (Al) ~1.1% > Others ~0.5%. - **Earth's Crust Composition**: **Oxygen (O) ~46.6%** > **Silicon (Si) ~27.7%** > **Aluminium (Al) ~8.1%** > **Iron (Fe) ~5.0%** > Calcium ~3.6% > Sodium ~2.8% > Potassium ~2.6% > Magnesium ~2.1%. - **Mean Density of Earth**: **5.518 g/cm³** (approx. **5.52**, corresponding to density of water; highest among all solar system planets).

Oceanic crust is about **5–10 km** thick. Continental crust is about **30–40 km** and thicker under young mountains. The rigid **lithosphere** (crust + uppermost mantle) rides on the weaker **asthenosphere**. The outer core is **liquid** iron–nickel. That liquid shell generates Earth's magnetic field and **stops S-waves**. The inner core is **solid** despite the heat because pressure is extreme.

Shadow	Note
S-wave	None beyond ~**103°** from epicentre (liquid outer core)
P-wave	Weak/absent ~**103–142°** (refraction in the core)

P-waves travel through solids, liquids and gases. **S-waves** travel through **solids only**. Beyond about **103°** from the epicentre, direct S-waves vanish because the outer core cannot transmit shear. Direct P-waves are missing in a belt about **103–142°** because they are strongly refracted at the mantle–core boundary. Beyond about **142°**, P-waves reappear after crossing the core. Surface waves stay near the surface and do most of the damage.

N.14 Continental Drift

Alfred Wegener proposed continental drift in **1912**.

- He argued that the continents once formed a single supercontinent called **Pangaea**.
- A vast ocean called **Panthalassa** surrounded Pangaea.
- Pangaea later split into northern **Laurasia** and southern **Gondwana**.
- Evidence includes the **jigsaw fit** of coasts (especially South America and Africa), matching **fossils** such as **Glossopteris** and **Mesosaurus**, matching rock belts, and glacial till deposits on now-separate continents.
- Wegener could describe the pattern but lacked a convincing **mechanism** for how continents moved.

How the idea grew after Wegener

- **Arthur Holmes** (~1928–29) suggested **mantle convection** as a driving force.
- **Harry Hess** (1960s) developed **sea-floor spreading** at mid-ocean ridges, with paleomagnetic stripes as support.
- Modern **plate tectonics** treats the lithosphere as moving plates. India was once part of **Gondwana**.
- New crust is added by **sea-floor spreading**. Old crust is destroyed by **subduction**.

Later piece	Fact
Mantle convection	**Holmes** (~1928–29)
Sea-floor spreading	**Harry Hess** (1960s); paleomagnetic stripes on MOR
Plates	Lithosphere pieces; India was in **Gondwana**

Geological-time pairs often asked with this map

- **Pleistocene** is the classic “Great Ice Age” framing in coaching notes.
- The **Little Ice Age** is often dated about **1300–1870 AD**, with colder winters in Europe and North America.
- The **Jurassic** is the age of dinosaurs. Indian remains notes include **Raiyoli (Gujarat)** and **Bara Simla / Narmada (नर्मदा) (Jabalpur)**.
- The **Cocos plate** sits between Central America and the Pacific plate.

N.15 Plate Tectonic Theory

Plate tectonics explains how rigid **lithosphere plates** move over the weaker **asthenosphere**. Mantle **convection**, slab pull and ridge push drive that motion. The theory absorbed Wegener’s drift idea and added a mechanism through **sea-floor spreading** and paleomagnetic stripes.

Seven major plates are the Pacific, North American, South American, Eurasian, African, Indo-Australian and Antarctic plates. Important smaller plates include **Nazca**, **Cocos** and the **Philippine** plate. The **Pacific** plate is the largest.

Boundary	Motion	Result
Divergent	Apart	MOR, rift (East African), new crust
Convergent	Together	Trenches, fold mountains, volcanoes, quakes
Transform	Slide	San Andreas; quakes, little crust created

At a **divergent** boundary, plates pull apart and new oceanic crust forms along a mid-ocean ridge or a continental rift. At a **convergent** boundary, plates collide. Oceanic crust may **subduct** into a trench, or two continents may collide and raise fold mountains. At a **transform** boundary, plates slide past each other and release energy as earthquakes with little crust created or destroyed.

The **Nazca** plate sinks under South America and builds the **Andes** beside the Peru–Chile trench. A **Benioff** zone is the dipping plane of earthquakes inside a subducting slab. India was once part of **Gondwana**. Its collision with Eurasia still thickens and raises the **Himalaya** (हिमालय).

N.16 Fold Mountains vs Fault (Block) Mountains

Mountain belts form in more than one way. **Fold mountains** rise where horizontal **compression** crumples thick sedimentary piles into anticlines, synclines and nappes. **Fault** or **block** mountains rise where the crust is stretched or broken into upthrown **horsts** and downthrown **grabens**.

	Fold	Fault / block / rift
Force	**Compression**	Tension / normal faulting
Form	Parallel ranges, nappes	Uplifted blocks (horst) + rifts (graben)
Examples	**Himalaya, Alps, Andes, Rockies, Atlas, Urals, Appalachians**	**Vosges, Black Forest, Sierra Nevada**, Rhine graben; some peninsular blocks

The **Himalaya**, Alps, Andes and Rockies are classic young fold belts on convergent margins. The **Vosges** and **Black Forest** face each other across the Rhine **graben** as paired block mountains. **Sierra Nevada** is another major fault-block range. Do not swap the pairs: Himalaya is **fold**, not fault; Vosges is **block**, not fold.

N.16A Rock Types & Rock Cycle

Petrology is the study of rocks. A rock is a natural solid aggregate of one or more minerals. Crust rocks fall into three families that recycle into each other.

Family	How it forms	Key traits
Igneous (primary)	Cooling of **magma / lava**	No fossils; not layered like sediment

Family	How it forms	Key traits
Sedimentary	Weathering → transport → deposition → **lithification**	Fossils common; occur in **layers (strata)**
Metamorphic	Any rock remade by heat + pressure + chemistry (**metamorphism**)	New fabric / minerals; forms deeper / hotter than surface conditions

- Igneous rock is hard for water to penetrate. It may be crystalline or non-crystalline.
- Sedimentary rock forms at or near Earth's surface.
- Metamorphic rock is any earlier rock remade by heat, pressure and chemistry (**metamorphism**).

Igneous split

Kind	Setting	Examples
Plutonic / intrusive	Magma cools **slowly deep** inside the crust	**Granite**, gabbro, pegmatite
Volcanic / extrusive	Lava cools at / near the surface	**Basalt**, volcanic breccia

Sedimentary types (coaching map)

Type	Examples
Mechanical (clastic)	Sandstone, shale, conglomerate; limestone often listed here in older books
Organic / carbonic	Coal; limestone / geyserite note
Chemical	Halite, potassium salts

Sandstone's flaky / layered look comes from deposition (often under water / low basins), not from desert heat alone.

Parent to metamorphic pairs

Parent	Metamorphic product
Granite	**Gneiss**
Basalt	Schist / amphibolite
Shale	**Slate**, then phyllite, then schist at higher grade
Limestone / chalk / dolomite	**Marble**
Sandstone	**Quartzite**
Conglomerate	Gneiss (listed in some GC sheets)
Gabbro	Serpentine (GC sheet)

- **Granite** is igneous — not metamorphic. **Gneiss / marble / quartzite / slate** are metamorphic.

- Sedimentary rocks can form from weathering of **all three** rock families, then cementation of fragments or chemical / organic growth in water.

N.17 Volcano Types

A volcano is a crustal rupture that lets lava, ash and gases escape from a magma chamber.

Note	Fact
Magma	Molten rock **below** the surface
Lava	Magma that reaches the surface (~600–1200°C); cools to igneous rock
Most abundant volcanic gas	**Water vapour (H_2O)** (जलवाष्प) — then CO_2, SO_2 and others
Acidic lava	High silica makes it sticky / viscous
Basic lava	Low silica makes it runny / fluid
Caldera	Large collapsed vent / crater
Geyser	Intermittent hot water + steam (e.g. Old Faithful, Yellowstone) — near magmatic heat
Fumarole / solfatara	Late-stage gas vents (a sulfurous vent is a solfatara); Katmai's "Valley of Ten Thousand Smokes" is the classic note
Type	Fact / example
------	----------
Shield	Low-silica, runny lava; classic Hawaii pair is **Mauna Loa / Mauna Kea**
Composite / strato	Sticky, explosive; **Fuji, Vesuvius, Etna, Pinatubo, St Helens, Rainier, Cotopaxi**
Cinder cone	Small, steep; **Paricutin** (Mexico)
Caldera	Collapse after blast; **Krakatoa / Krakatau** (Indonesia), Yellowstone
Fissure / flood basalt	**Deccan Trap** (दक्कन ट्रैप)

- **Mauna Loa** is Earth's largest volcano by bulk. It also had a strong **2022** eruption.

Status	Fact
Active	Open vent / recent eruptions — **Stromboli** ("lighthouse of the Mediterranean"), **Etna** (Sicily), **Kilauea** (often ranked among most active), **Mauna Loa**, **Barren Island** (India's only confirmed active — Andaman Sea), **Fuji**, **Vesuvius**, **Erebus** (Ross Island, Antarctica), **Cotopaxi** (Ecuador)
Highest active (world)	Coaching often cites **Ojos del Salado** (Andes, Chile–Argentina, ~6893 m). Among common options **Cotopaxi** may be the tallest listed
Dormant	Quiet but may revive — **Kilimanjaro** (Kibo dormant; Mawenzi / Shira extinct) and **Damavand** (Iran)

Status	Fact
Extinct	Unlikely to erupt again — Kulal (Kenya), Chimborazo (Ecuador), Popa (Myanmar), Koh-e-Sultan (Pakistan) note

Beyond the Rainier–Etna–Paricutin–Apo set, learn these volcano–country pairs for Match List practice.

Volcano / peak	Country	Teaching note
Sabancaya	**Peru**	Andean volcano
Colima	**Mexico**	Mexican volcano match
Merapi	**Indonesia**	Classic Indonesian volcano
Kinabalu	**Malaysia**	Peak often listed with volcano/country sets
Elburz (Alborz)	**Iran**	Iranian range / peak match
Aconcagua	**Argentina**	High peak; **not** treated as an active volcano in standard keys
Krakatoa / Krakatau	**Indonesia**	Famous caldera volcano

A volcanic cone is a **constructional** landform built by lava and ash, not a residual hill left by erosion.

Inline PYQ — UPPCS Prelims 2021, Q78

Match List-I with List-II and select the correct answer.

List-I (Volcano)	List-II (Country)
A. Rainier	1. Italy
B. Etna	2. Mexico
C. Paricutin	3. Philippines
D. Apo	4. USA

> Row order is **not** the answer — use the option codes.

A. 4, 2, 1, 3

B. 4, 1, 2, 3

C. 2, 1, 4, 3

D. 4, 3, 2, 1

Show answer

Ans: B — Rainier–USA; Etna–Italy; Paricutin–Mexico; Apo–Philippines.

N.17B Earthquakes · Seismic Waves · Tsunami · Scales

An earthquake is shaking from a sudden release of energy in the **lithosphere**. Natural earthquakes occur in the lithosphere.

Earthquake terms — Teaching Card

- The **focus** (or **hypocentre**) is the point of **initial rupture** inside the Earth — where the slip starts underground.
- The **epicentre** is the point on the **surface directly above** the focus — the surface spot coaching keys tie to felt effects and first observations.
- Memory pair: **focus = underground start**; **epicentre = surface above it**.
- A **seismograph / seismometer** detects and records the waves; a **seismogram** is the paper or digital **record** of that motion.

Term	Quick recall
Focus / hypocentre	Underground rupture point
Epicentre	Surface point above the focus
Seismograph / seismometer	Instrument that records waves
Seismogram	The wave record itself

Three wave families — remember **P** (Primary), **S** (Secondary), and **surface** (Love / Rayleigh).

Wave	Nature	Travels through	Note
P (primary)	Longitudinal / sound-like; **fastest**; arrives first	Solids, liquids, gases	Body wave
S (secondary)	Transverse / light-like; slower than P	**Solids only**	Body wave
Surface	Slowest; Love + Rayleigh; **most damaging** at surface	Near surface	Love waves are the faster surface type

- **S-waves** are stopped by the liquid outer core.
- Surface waves do the worst damage at the ground.

Shadow zones (NCERT): Seismographs within about **103°** of the epicentre record both P and S waves. **S-waves** are not recorded beyond about **103°** because the liquid outer core stops shear waves. **P-waves** are missing in a belt about **103–142°** because they are refracted at the mantle–core boundary; they reappear beyond about **142°**. A claim that both P and S arrive at *any* distance beyond 105° is **false**.

Magnitude vs intensity (मर्कली)

Scale	Measures	Note
Richter (C.F. Richter, **1935**)	**Magnitude** ≈ energy released ($\log_{10}$ amplitude)	+1 magnitude ≈ **10×** amplitude; ≈ **~32×** energy (not 100×)
Mercalli	**Intensity** (felt damage / effects)	Different from magnitude

Prefer **Mercalli** for intensity and **Richter** for magnitude when both scales appear in the options. Some older keys loosely treat Richter as an intensity scale when Mercalli is missing — do not learn that swap as the main rule.

Tsunami comes from Japanese *tsu* (harbour) and *nami* (wave). Undersea quakes, volcanic eruptions and underwater explosions can trigger it. A tsunami can cross whole ocean basins.

Event pair	Fact
26 Dec 2004 Indian Ocean	Sumatra subduction quake. Waves reached as far as Africa. Deaths were about **2–3 lakh**, not "50 lakh". Bangladesh saw relatively few casualties because the fault trended roughly N–S while main wave energy ran roughly E–W
11 Mar 2011 Japan	A great quake and tsunami damaged **Fukushima** Daiichi and caused a nuclear disaster

India: NW subcontinent is quake-prone because it sits on the **Indian–Eurasian plate convergence**. India's Himalayan / mid-continental arm links to the Alpine–Himalayan seismic belt. It is not limited to the Pacific Ring alone.

Plate-tectonics pair: Sea-floor spreading (Hess and others) underpins plate tectonics. Pinning "plate tectonics theory" on Hess alone in 1967 is too narrow. The statement that the theory is based on sea-floor spreading stays true.

N.18 Major Earthquake & Volcano Belts

Most earthquakes and volcanoes cluster along **plate boundaries**, not randomly over continents. Four belts carry almost all of the world's seismic and volcanic activity.

Belt	Note
Circum-Pacific / Ring of Fire	Pacific rim arc (Chile–California–Alaska–Japan–Philippines–New Zealand)
Mid-Atlantic	Iceland, Azores — divergent; quakes + volcanoes
Alpine–Himalayan / Mid-Continental	Cape Verde–Atlantic–Mediterranean–Alps–Caucasus–**Himalaya**; India's main seismic framing
East African Rift	**Kilimanjaro, Kenya — not** Ring of Fire ()
Cocos plate	Between **Central America** and the **Pacific** plate

The **Circum-Pacific Ring of Fire** follows subduction around the Pacific basin. It holds most of the world's volcanoes and about nine-tenths of recorded earthquakes. The name is about fire from magma, **not** forest fires. The **Mid-Atlantic** belt follows the divergent mid-ocean ridge and includes Iceland and the Azores. The **Alpine–Himalayan** belt runs from the Mediterranean through the Alps and Caucasus into the **Himalaya**. It is India's main inland seismic framing because the Indian plate still pushes into Eurasia. The **East African Rift** is a divergent continental rift with volcanoes such as **Kilimanjaro** and Kenya. It is **not** part of the Pacific Ring of Fire. The **Cocos** plate lies between Central America and the Pacific plate and feeds Central American subduction.

India's official seismic zoning uses **Zones II–V**, with Zone V the highest hazard. The **Prime Minister** chairs the NDMA (आपदा प्राधिकरण).
